ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

SYSTEMS ANALYSIS AND DESIGN

PRACTICE EXAM · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Systems Analysis and Design practice exam.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Answers must relate to the case study where asked. Accept any standard notation for diagrams and keys. Accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 D ✓ 1.1.3 B ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 B ✓ 1.1.7 D ✓ 1.1.8 A ✓ 1.1.9 C ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 I ✓ 1.3.2 A ✓ 1.3.3 F ✓ 1.3.4 B ✓ 1.3.5 C ✓ 1.3.6 E ✓ 1.3.7 H ✓ 1.3.8 J ✓ 1.3.9 G ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Approach: Waterfall – sequential phases, each finished before the next ✓; Agile – iterative, small working parts in short cycles ✓. Client involvement: Waterfall – mainly at the start and end ✓; Agile – continuous feedback throughout ✓. Changes: Waterfall – difficult and costly once a phase is complete ✓; Agile – welcomed and handled in the next sprint ✓. Recommendation: Agile ✓ – the booking, tracking and mobile features can be delivered and tested step by step and requirements may change ✓ (accept Waterfall with a sound reason) (8)

2.2 Functional (any THREE) ✓✓✓: customers book a repair online; customers track the status of a job; manager assigns technicians; technicians record parts and hours; system generates invoices; reports on jobs and income. Non-functional (any TWO) ✓✓: works on phones (responsive); secure login / data protection; available 24/7; pages load within 3 seconds; easy to use (5)

2.3 E.g. "As a technician, I want to update the job status and parts used on my phone, so that the office can invoice the customer immediately." ✓ (correct format) ✓ (relevant) (2)

[15]

QUESTION 3

3.1 Actors (any THREE) ✓✓✓: Customer, Receptionist, Technician, Manager. Use cases (any SIX, correctly linked) ✓✓✓✓✓✓: Book repair, Track repair, Register / log in, Assign technician, Update job, Record parts used, Generate invoice, View reports, Make payment. System boundary rectangle labelled "QuickFix System" ✓ (10)

3.2 Three classes with two attributes and one method each ✓✓✓ (e.g. Customer: custNo, name; bookRepair(). Job: jobNo, status; updateStatus(). Technician: techID, name; viewJobs()). Customer 1 — 0..* Job ✓; Technician 1 — 0..* Job ✓; correct notation ✓ (6)

3.3 It shows the flow of activities / steps in a process ✓, including decisions and parallel actions, e.g. the steps from booking to invoicing ✓. (2)

[18]

QUESTION 4

4.1.1 Any TWO: update anomaly – a customer's phone number is stored many times and must be changed in every row ✓; insertion anomaly – a new part or technician cannot be added without a job ✓; deletion anomaly – deleting a job may delete the only record of a customer; repeating groups (several parts in one field) (2)

4.1.2 ✓ per correct table (with fields), ✓ for correct primary keys, ✓ for correct foreign keys:

Table	Fields
CUSTOMER	CustNo, CustName, CustPhone ✓✓
TECHNICIAN	TechID, TechName ✓✓
JOB	JobNo, Date, CustNo*, TechID* ✓✓
PART	PartNo, PartDesc, Price ✓✓
JOB_PART	JobNo*, PartNo* (composite key), Qty ✓✓

(10)

4.1.3 $(1 \times \text{R}450) + (2 \times \text{R}120)$ ✓ $= \text{R}690$ ✓ (2)

4.2 Partial dependency: a non-key field depends on only part of a composite key ✓, e.g. PartDesc depends only on PartNo, not on (JobNo, PartNo) ✓. Transitive dependency: a non-key field depends on another non-key field ✓, e.g. CustName depends on CustNo, which depends on JobNo ✓. (4)

[18]

QUESTION 5

5.1 Unit testing – each module (e.g. booking form) tested on its own ✓✓; integration testing – modules tested together (booking → assignment → invoice) ✓✓; system testing – the whole system tested against the requirements, including performance and security ✓✓; acceptance testing – QuickFix staff and customers test it with real data before sign-off ✓✓ (8)

5.2 Any THREE ✓✓✓: HTTPS / SSL encryption; strong passwords and two-factor authentication; role-based access (technicians see only their jobs); encrypt the database; regular backups; firewall; audit trails; regular security updates. POPIA link ✓: personal information must be kept secure and only used for the purpose it was collected for / customers must consent (4)

5.3 Any TWO explained ✓✓ each: user training (receptionist, technicians, manager); data conversion (capture existing customers and jobs); hardware and network setup (phones, Internet, hosting); documentation (user manuals); support / help desk after go-live; timing (implement in a quiet period) (4)

5.4 Any THREE: reliable / accurate ✓; secure ✓; easy to use ✓; fast (efficient); maintainable; meets user requirements; scalable (3)

[19]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE